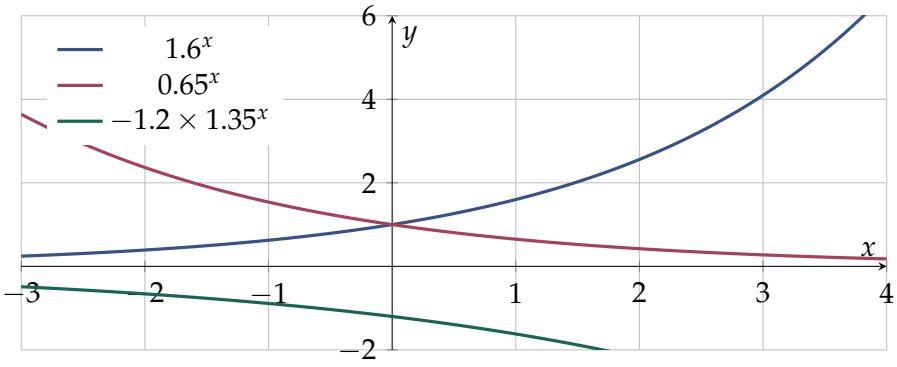
CHAPTER 2

Exponential Functions

Skills: model a continuous change, switch between representations, calculate, interpret

2.1 Core Course Notes

For $a > 0$, the function $x \mapsto a^x$ extends the integer powers of a to all real numbers. It satisfies $a^0 = 1$ and $a^{-x} = \dfrac{1}{a^x}$.



Key point. If $a > 1$, $x \mapsto a^x$ is strictly increasing. If $0 < a < 1$, it is strictly decreasing. For $f(x) = ka^x$, multiplying by $k < 0$ reverses the direction of variation. In every case, $a^x > 0$.

For $a > 0$ and all real numbers x, y:

$$a^{x+y} = a^x a^y, \qquad a^{x-y} = \frac{a^x}{a^y}, \qquad (a^x)^y = a^{xy}.$$

Watch out. In general, $a^x + a^y \neq a^{x+y}$. The rules apply to products and quotients, not to sums.

2.1.1 Changes and average rate

A quantity with initial value V_0 undergoing a constant relative change with multiplier $q > 0$ is modelled by $V(t) = V_0 q^t$. If an overall multiplier $Q > 0$ results from n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $\tau = q - 1$.

Example 1. A company's turnover goes from 80,000 to 95,000 euros in four years. The overall multiplier is $Q = \dfrac{95{,}000}{80{,}000} = 1.1875$. The average annual multiplier is $q = 1.1875^{\frac{1}{4}} \approx 1.0439$: the average annual rate is therefore about 4.39%.

2.1.2 Solving and interpreting

When both sides can be written with the same base, we use injectivity: for $a > 0$ and $a \neq 1$, $a^u = a^v$ is equivalent to $u = v$. Otherwise, a graphical or numerical solution gives an approximate value; the common logarithm (base 10), studied later, will give a formula.

Session 7 — Definition and direction of variation of a^x

Exponential Functions

6 Oct 2026 • B: 2 h – C: 1 h

Objectives.

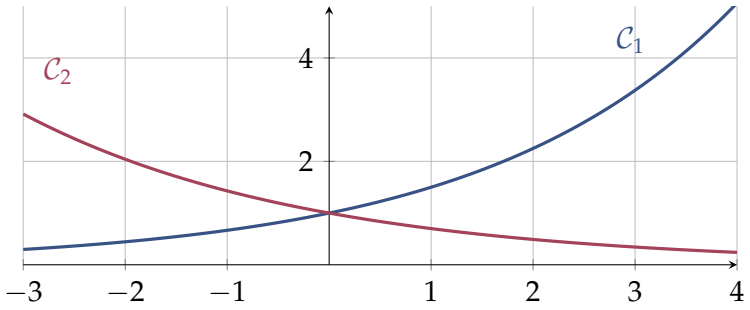
- Make sense of a^x.
- Link the base to the direction of variation.
- Read curves.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. $1 + \dfrac{10}{100} = 1.10$
2. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
3. Calculate $2^4 \times 2^3$. 2^7
4. Write 5^{-2} as a fraction. $\dfrac{1}{25}$
5. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: make sense of a^x and link the base to the direction of variation.

Activity 1. Using a calculator, complete a table of values of 1.5^x and 0.7^x for $x = -2, -1, 0, 1, 2, 3$. Plot the points, conjecture the variations and identify the point common to both curves.



Solution. $a^0 = 1$, so the curves pass through $(0, 1)$. The first increases and the second decreases. Have students say aloud which piece of data triggers the choice of method.

Written record. For $a > 0$, a^x is defined for every real number x. The function is increasing if $a > 1$ and decreasing if $0 < a < 1$.

Example 2. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $a^0 = 1$, so the curves pass through $(0, 1)$. The first increases and the second decreases.

Exercise 2.1 · essential.

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1.04^x$ and $g(x) = -4 \times 0.8^x$.
3. An initial value of 1350 increases by 4% per year. Give the model and calculate the value after 6 years.

Solution. 3^4 and 2^{3x}. f is increasing because $1.04 > 1$ and $4 > 0$; 0.8^x is decreasing and the coefficient -4 is negative, so g is increasing. The model is $V(t) = 1350 \times \left(1 + \dfrac{4}{100}\right)^t$. After 6 years, $V(6) = 1350 \times 1.04^6 \approx 1{,}708.18$.

Application. On 1 January 2024, the capacity of the solar panels installed in a region is 240 megawatts (MW). This capacity is estimated to grow by 12% per year. For every natural number n, let u_n be the installed capacity, in MW, on 1 January of the year $2024 + n$; we assume that (u_n) is a geometric sequence.

1. Give the common ratio of the sequence (u_n) and express u_n in terms of n.
2. We want to estimate the installed capacity at any date. Suggest an expression for a function f giving the installed capacity x years after 1 January 2024, where x is a positive real number.
3. Estimate, to the nearest 0.1 MW, the installed capacity on 1 January 2026, on 1 April 2027 and on 1 July 2029. A quarter is taken as 0.25 year.

Solution. **Target:** Extend a geometric sequence to a function.

1. A 12% increase corresponds to the multiplier $1 + \dfrac{12}{100} = 1.12$. The common ratio is $q = 1.12$ and $u_n = 240 \times 1.12^n$.
2. We extend the sequence to positive real numbers: $f(x) = 240 \times 1.12^x$ (indeed $f(n) = u_n$ for every integer n).
3. - 1 January 2026: $x = 2$ and $f(2) = 240 \times 1.12^2 = 301.056 \approx 301.1$ MW.
   - 1 April 2027: 3 years and one quarter, so $x = 3.25$; $f(3.25) = 240 \times 1.12^{3.25} \approx 346.9$ MW.
   - 1 July 2029: 5 and a half years, so $x = 5.5$; $f(5.5) = 240 \times 1.12^{5.5} \approx 447.6$ MW.

   According to this model, the installed capacity would be about 301.1 MW, 346.9 MW and 447.6 MW on these three dates.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 155 and the rate or ratio by 3. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 8 — Powers and algebraic rules

Exponential Functions

8 Oct 2026 • B: 1 h – C: 2 h

Objectives.

- Transform powers.
- Use negative exponents.
- Spot a rule that does not apply.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 4 - x + 2$. $2x + 6$
2. Calculate $2^1 \times 2^3$. 2^4
3. Write 5^{-2} as a fraction. $\dfrac{1}{25}$
4. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
5. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: transform powers and use negative exponents.

Activity 2. Simplify $2^3 \times 2^{-5}$, $5^{x+2} \div 5^x$ and $(3^x)^2$. Then invent a numerical counterexample to $a^x + a^y = a^{x+y}$.

Solution. We obtain $2^{-2} = \dfrac{1}{4}$, $5^2 = 25$ and 3^{2x}. With $a = 2$ and $x = y = 1$ we have $2 + 2 = 4$: this is an accidental equality, which proves nothing. With $a = 3$ and $x = y = 1$: $3 + 3 = 6 \neq 9 = 3^2$, which is a counterexample. Have students say aloud which piece of data triggers the choice of method.

Written record. $a^{x+y} = a^x a^y$, $a^{x-y} = \dfrac{a^x}{a^y}$ and $(a^x)^y = a^{xy}$. These rules require the same positive base.

Example 3. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. We obtain $2^{-2} = \dfrac{1}{4}$, $5^2 = 25$ and 3^{2x}. With $a = 2$ and $x = y = 1$ we have $2 + 2 = 4$: this is an accidental equality, which proves nothing. With $a = 3$ and $x = y = 1$: $3 + 3 = 6 \neq 9 = 3^2$, which is a counterexample.

Exercise 2.2 · essential.

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1.05^x$ and $g(x) = -5 \times 0.8^x$.
3. An initial value of 1400 increases by 5% per year. Give the model and calculate the value after 6 years.

Solution. 3^5 and 2^{3x}. f is increasing because $1.05 > 1$ and $5 > 0$; 0.8^x is decreasing and the coefficient -5 is negative, so g is increasing. The model is $V(t) = 1400 \times \left(1 + \dfrac{5}{100}\right)^t$. After 6 years, $V(6) = 1400 \times 1.05^6 \approx 1{,}876.13$.

Application. Write each of the following numbers as a single power of 3:

$$a = 3^{-2} \times 3^6; \qquad b = \left(3^{1.5}\right)^4; \qquad c = 27 \times 9^{-2}; \qquad d = \frac{3^8}{9^3}.$$

Solution. **Target:** Algebraic rules for powers.

- $a = 3^{-2+6} = 3^4$ (rule $a^x a^y = a^{x+y}$).
- $b = 3^{1.5\times 4} = 3^6$ (rule $(a^x)^y = a^{xy}$).
- $27 = 3^3$ and $9 = 3^2$, so $9^{-2} = \left(3^2\right)^{-2} = 3^{-4}$; hence $c = 3^3 \times 3^{-4} = 3^{-1}$.
- $9^3 = \left(3^2\right)^3 = 3^6$, so $d = \dfrac{3^8}{3^6} = 3^{8-6} = 3^2$.

This can be checked on a calculator, for example $c = \dfrac{27}{81} = \dfrac{1}{3} = 3^{-1}$ and $d = \dfrac{6{,}561}{729} = 9 = 3^2$.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 160 and the rate or ratio by 4. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 9 — Functions $x \mapsto ka^x$: variation and graph

Exponential Functions

13 Oct 2026 • B: 2 h – C: 1 h

Objectives.

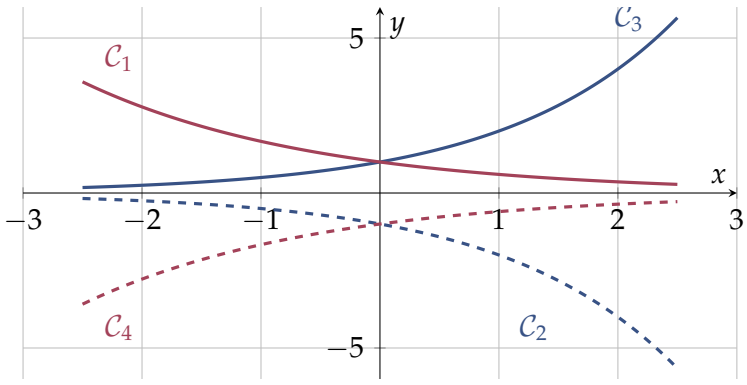
- Study ka^x.
- Combine sign and variation.
- Link formula and graph.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 100. 20
2. Write 5^{-2} as a fraction. $\dfrac{1}{25}$
3. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
4. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
5. Calculate $2^2 \times 2^3$. 2^5

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: study ka^x and combine sign and variation.

Activity 3. The four curves $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$ and $\mathcal{C}_4$ below represent, in an unknown order, the functions 2^x, 0.6^x, -2^x and -0.6^x. Match each function to its curve. Justify each choice by the sign, the y-intercept and the direction of variation.



Solution. All of them have an absolute value equal to 1 at 0. The sign distinguishes the two pairs; the base and the sign of the coefficient determine the variation. Matching: $\mathcal{C}_3$ is 2^x (positive, increasing, passes through $(0,1)$); $\mathcal{C}_1$ is 0.6^x (positive, decreasing, passes through $(0,1)$); $\mathcal{C}_2$ is -2^x (negative, decreasing, passes through $(0,-1)$); $\mathcal{C}_4$ is -0.6^x (negative, increasing, passes through $(0,-1)$). Have students say aloud which piece of data triggers the choice of method.

Written record. For $f(x) = ka^x$, the sign is that of k. If $k < 0$, multiplying by k reverses the direction of variation of a^x.

Example 4. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. All of them have an absolute value equal to 1 at 0. The sign distinguishes the two pairs; the base and the sign of the coefficient determine the variation. Matching: $\mathcal{C}_3$ is 2^x (positive, increasing, passes through $(0,1)$); $\mathcal{C}_1$ is 0.6^x (positive, decreasing, passes through $(0,1)$); $\mathcal{C}_2$ is -2^x (negative, decreasing, passes through $(0,-1)$); $\mathcal{C}_4$ is -0.6^x (negative, increasing, passes through $(0,-1)$).

Exercise 2.3 · essential.

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1.06^x$ and $g(x) = -6 \times 0.8^x$.
3. An initial value of 1450 increases by 6% per year. Give the model and calculate the value after 6 years.

Solution. 3^6 and 2^{3x}. f is increasing because $1.06 > 1$ and $6 > 0$; 0.8^x is decreasing and the coefficient -6 is negative, so g is increasing. The model is $V(t) = 1450 \times \left(1 + \dfrac{6}{100}\right)^t$. After 6 years, $V(6) = 1450 \times 1.06^6 \approx 2{,}056.85$.

Application. Determine the direction of variation on $\mathbb{R}$ of the functions defined by:

$$f : t \mapsto 3.5 \times 0.82^t; \qquad g : t \mapsto -0.6 \times 1.15^t; \qquad h : t \mapsto -4 \times 0.7^t.$$

Solution. **Target:** Direction of variation of $t \mapsto ka^t$.

- $0 < 0.82 < 1$, so $t \mapsto 0.82^t$ is strictly decreasing; the coefficient 3.5 is positive and keeps the direction of variation: f is strictly decreasing.
- $1.15 > 1$, so $t \mapsto 1.15^t$ is strictly increasing; the coefficient -0.6 is negative and reverses the direction of variation: g is strictly decreasing.
- $0 < 0.7 < 1$, so $t \mapsto 0.7^t$ is strictly decreasing; the coefficient -4 is negative and reverses the direction of variation: h is strictly increasing.

Numerical check: $h(0) = -4$, $h(1) = -2.8$ and $h(2) = -1.96$: the values increase.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 165 and the rate or ratio by 5. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 10 — Average rate of change

Exponential Functions

15 Oct 2026 • B: 1 h – C: 2 h

Objectives.

- Calculate an average rate.
- Distinguish overall and average rates.
- Interpret an exponent $\frac{1}{n}$.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 25% increase. $1 + \dfrac{25}{100} = 1.25$
2. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
3. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
4. Calculate $2^3 \times 2^3$. 2^6
5. Write 5^{-2} as a fraction. $\dfrac{1}{25}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: calculate an average rate and distinguish overall and average rates.

Activity 4. The price of a service goes from 18 to 23.50 euros in five years. Calculate the overall multiplier, then the average annual multiplier and the average annual rate. Check by applying the change five times in a row.

Solution. $Q = \dfrac{23.5}{18} \approx 1.3056$, $q = Q^{\frac{1}{5}} \approx 1.0548$, that is, about 5.48% per year. Have students say aloud which piece of data triggers the choice of method.

Written record. If Q is the overall multiplier of n identical changes, the average multiplier is $q = Q^{\frac{1}{n}}$ and the average rate is $q - 1$.

Example 5. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $Q = \dfrac{23.5}{18} \approx 1.3056$, $q = Q^{\frac{1}{5}} \approx 1.0548$, that is, about 5.48% per year.

Exercise 2.4 · essential.

1. Simplify $3^{x+2} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 2 \times 1.02^x$ and $g(x) = -2 \times 0.8^x$.
3. An initial value of 1500 increases by 2% per year. Give the model and calculate the value after 6 years.

Solution. 3^2 and 2^{3x}. f is increasing because $1.02 > 1$ and $2 > 0$; 0.8^x is decreasing and the coefficient -2 is negative, so g is increasing. The model is $V(t) = 1500 \times \left(1 + \dfrac{2}{100}\right)^t$. After 6 years, $V(6) = 1500 \times 1.02^6 \approx 1{,}689.24$.

Application. Over five years, a factory has reduced its water consumption by 27%.

a. Give the overall multiplier over the five years.

b. Calculate the average annual rate of change of the water consumption, to the nearest 0.1%.

c. Check the result by applying this rounded average multiplier five times.

Solution. **Target:** Average annual rate of change.

a. A 27% decrease corresponds to the overall multiplier $Q = 1 - \dfrac{27}{100} = 0.73$.

b. If q is the average annual multiplier, then $q^5 = 0.73$, so $q = 0.73^{\frac{1}{5}} \approx 0.9390$. The average annual rate is $t = q - 1 \approx -0.061$: water consumption fell on average by about 6.1% per year.

c. $0.939^5 \approx 0.7300$, which does correspond to an overall decrease of about 27%.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 170 and the rate or ratio by 6. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 11 — Reading graphical representations

Exponential Functions

3 Nov 2026 • B: 2 h – C: 1 h

Objectives.

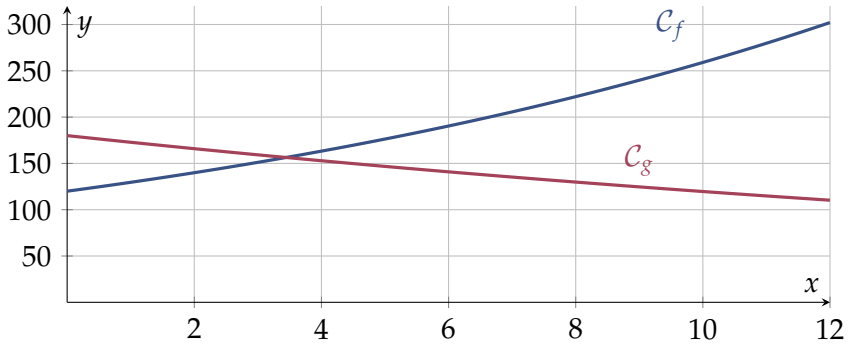
- Read outputs and find the inputs that give them.
- Compare two models.
- Estimate graphically.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 7 - x + 2$. $2x + 9$
2. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
3. Calculate $2^4 \times 2^3$. 2^7
4. Write 5^{-2} as a fraction. $\frac{1}{25}$
5. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: read outputs and find the inputs that give them and compare two models.

Activity 5. Two commercial offers are modelled by $f(x) = 120 \times 1.08^x$ and $g(x) = 180 \times 0.96^x$. Draw the curves on $[0, 12]$, read off their point of intersection and interpret the regions where each value is the greater.



Solution. Reading the graph places the intersection near $x \approx 3.4$ (calculated value: $\frac{\log 1.5}{\log 1.125} \approx 3.44$), for an ordinate close to $180 \times 0.96^{3,44} \approx 156$. Before this point, g is greater than f; after it, f is greater than g. An approximate value is expected. Have students say aloud which piece of data triggers the choice of method.

Written record. Reading $f(x)$ means going from x to the curve and then to the y-axis. Finding an input for a given y means going from y to the curve and then to the x-axis.

Example 6. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

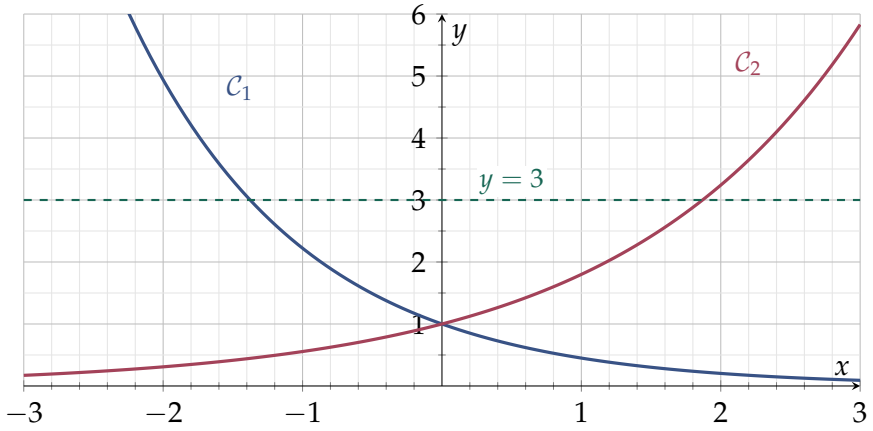
Solution. Reading the graph places the intersection near $x \approx 3.4$ (calculated value: $\dfrac{\log 1.5}{\log 1.125} \approx 3.44$), for an ordinate close to $180 \times 0.96^{3,44} \approx 156$. Before this point, g is greater than f; after it, f is greater than g. An approximate value is expected.

Exercise 2.5 · essential.

1. Simplify $3^{x+3} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 3 \times 1.03^x$ and $g(x) = -3 \times 0.8^x$.
3. An initial value of 1550 increases by 3% per year. Give the model and calculate the value after 6 years.

Solution. 3^3 and 2^{3x}. f is increasing because $1.03 > 1$ and $3 > 0$; 0.8^x is decreasing and the coefficient -3 is negative, so g is increasing. The model is $V(t) = 1550 \times \left(1 + \dfrac{3}{100}\right)^t$. After 6 years, $V(6) = 1550 \times 1.03^6 \approx 1{,}850.78$.

Application. The graph below shows, in a coordinate system, the functions f and g defined on $\mathbb{R}$ by $f(x) = 1.8^x$ and $g(x) = 0.45^x$, together with the line with equation $y = 3$.



1. Which curve represents the function f? Justify your choice.
2. Solve the equation $f(x) = 3$ graphically. Using a table of values on the calculator, give an approximate value of the solution to the nearest 0.01.
3. Determine graphically the set of real numbers x such that $f(x) \geqslant 3$, then the set of real numbers x such that $g(x) \geqslant 3$.

Solution. **Target:** Reading graphs of a^x curves.

1. $1.8 > 1$, so f is strictly increasing; $0 < 0.45 < 1$, so g is strictly decreasing. The increasing curve $\mathcal{C}_2$ represents f (and $\mathcal{C}_1$ represents g). Both curves pass through the point $(0, 1)$.
2. The curve $\mathcal{C}_2$ meets the line $y = 3$ at a point whose x-coordinate reads close to 1.9. With the calculator: $1.8^{1.86} \approx 2.984$ and $1.8^{1.87} \approx 3.002$, so the solution lies between 1.86 and 1.87; the value 1.87 gives the image closest to 3. The solution is $x \approx 1.87$.
3. $f(x) \geqslant 3$ when $\mathcal{C}_2$ is above the line $y = 3$, that is for $x \geqslant 1.87$ approximately: $[1.87, +\infty)$ (approximate value of the endpoint).

   g is decreasing and $\mathcal{C}_1$ meets the line $y = 3$ at an abscissa close to -1.4 (calculator: $0.45^{-1.38} \approx 3.010$ and $0.45^{-1.37} \approx 2.986$, so $x \approx -1.38$). Hence $g(x) \geqslant 3$ for $x \leqslant -1.38$ approximately: $(-\infty, -1.38]$ (approximate value of the endpoint).

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 175 and the rate or ratio by 7. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 12 — **Simple exponential equations**

Exponential Functions — 5 Nov 2026 • B: 1 h – C: 2 h

Objectives.

- Solve using a common base.
- Approximate a solution.
- Check plausibility.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 160. 8
2. Calculate $2^1 \times 2^3$. 2^4
3. Write 5^{-2} as a fraction. $\frac{1}{25}$
4. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
5. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: solve using a common base and approximate a solution.

Activity 6. Solve $4^x = 64$, $9^x = \frac{1}{3}$ and $1.12^x = 2$: first using a common base when possible, then numerically for the last one. Bracket the last solution between two integers.

Solution. $x = 3$; $9^x = 3^{-1}$ so $2x = -1$ and $x = -\frac{1}{2}$; finally $x \approx 6.12$, between 6 and 7. Have students say aloud which piece of data triggers the choice of method.

Written record. For $a > 0$ and $a \neq 1$, to solve $a^x = b$: use a common base when possible; otherwise bracket the solution or use a numerical tool. Check that $b > 0$.

Example 7. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $x = 3$; $9^x = 3^{-1}$ so $2x = -1$ and $x = -\frac{1}{2}$; finally $x \approx 6.12$, between 6 and 7.

Exercise 2.6 · essential.

1. Simplify $3^{x+4} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 4 \times 1.04^x$ and $g(x) = -4 \times 0.8^x$.

3. An initial value of 1600 increases by 4% per year. Give the model and calculate the value after 6 years.

Solution. 3^4 and 2^{3x}. f is increasing because $1.04 > 1$ and $4 > 0$; 0.8^x is decreasing and the coefficient -4 is negative, so g is increasing. The model is $V(t) = 1600 \times \left(1 + \dfrac{4}{100}\right)^t$. After 6 years, $V(6) = 1600 \times 1.04^6 \approx 2{,}024.51$.

Application. Solve the following inequalities on $\mathbb{R}$.

a. $0.7^x \leqslant 0.7^{-1}$

b. $2.5^{x+1} > 2.5^4$

c. $0.5^x \geqslant 0.25$

Solution. **Target:** Inequalities with the same base.

a. $0 < 0.7 < 1$: the function $x \mapsto 0.7^x$ is strictly decreasing, so it reverses the order. Hence $0.7^x \leqslant 0.7^{-1} \iff x \geqslant -1$. Solution set: $[-1, +\infty)$.

b. $2.5 > 1$: the function $x \mapsto 2.5^x$ is strictly increasing, so it preserves the order. Hence $2.5^{x+1} > 2.5^4 \iff x + 1 > 4 \iff x > 3$. Solution set: $(3, +\infty)$.

c. $0.25 = 0.5^2$, so the inequality reads $0.5^x \geqslant 0.5^2$. Since $x \mapsto 0.5^x$ is strictly decreasing: $x \leqslant 2$. Solution set: $(-\infty, 2]$.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 180 and the rate or ratio by 2. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 13 — Modelling with an exponential function

Exponential Functions

10 Nov 2026 • B: 2 h – C: 1 h

Objectives.

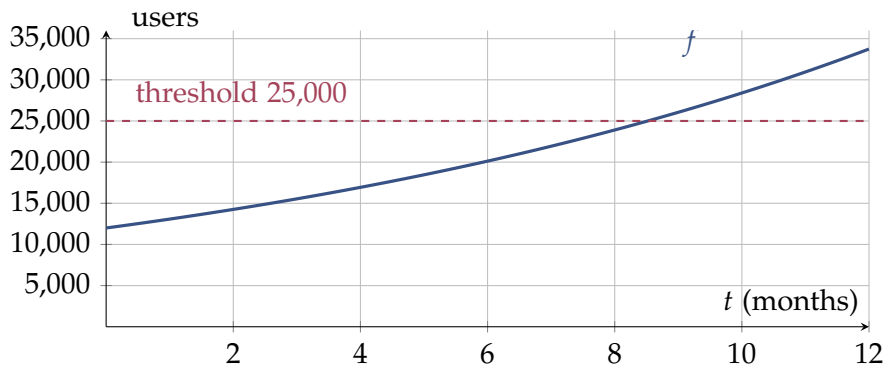
- Build an exponential model.
- Identify the parameters.
- Predict and interpret.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. $1 + \dfrac{10}{100} = 1.10$
2. Write 5^{-2} as a fraction. $\dfrac{1}{25}$
3. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
4. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
5. Calculate $2^2 \times 2^3$. 2^5

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: build an exponential model and identify the parameters.

Activity 7. An app has 12,000 users at launch and its usage grows by 9% per month. Build $f(t)$, calculate $f(6)$, then determine numerically when it will exceed 25,000 users.



Solution. $f(t) = 12{,}000 \times 1.09^t$, $f(6) \approx 20{,}125$. The threshold is exceeded as soon as $t = 9$, that is, 9 months after the launch, because $f(8) \approx 23{,}911 < 25{,}000 < f(9) \approx 26{,}063$. Have students say aloud which piece of data triggers the choice of method.

Written record. A constant relative change with multiplier q is modelled by $f(t) = V_0 q^t$. V_0 is the initial value and q the multiplier per unit of time.

Example 8. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $f(t) = 12{,}000 \times 1.09^t$, $f(6) \approx 20{,}125$. The threshold is exceeded as soon as $t = 9$, that is, 9 months after the launch, because $f(8) \approx 23{,}911 < 25{,}000 < f(9) \approx 26{,}063$.

Exercise 2.7 · essential.

1. Simplify $3^{x+5} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 5 \times 1.05^x$ and $g(x) = -5 \times 0.8^x$.
3. An initial value of 1650 increases by 5% per year. Give the model and calculate the value after 6 years.

Solution. 3^5 and 2^{3x}. f is increasing because $1.05 > 1$ and $5 > 0$; 0.8^x is decreasing and the coefficient -5 is negative, so g is increasing. The model is $V(t) = 1650 \times \left(1 + \dfrac{5}{100}\right)^t$. After 6 years, $V(6) = 1650 \times 1.05^6 \approx 2{,}211.16$.

Application. Just after an injection, the concentration of a drug in a patient's blood is 80 mg/L. This concentration then decreases by 15% every hour. This is an exponential decay.
It is modelled by a function C of the time t, in hours, such that $C(t) = ka^t$.

1. Determine the constants k and a.
2. Calculate the concentration after 3 h 30 min, to the nearest 0.1 mg/L.
3. The doctor plans a new injection as soon as the concentration drops below 50 mg/L. Is this already the case after 3 h 30 min? Answer in a sentence.

Solution. **Target:** Model with $f(t) = ka^t$ and compute.

1. $C(0) = k \times a^0 = k$, and $C(0) = 80$, so $k = 80$. A 15% decrease per hour corresponds to the multiplier $1 - \dfrac{15}{100} = 0.85$, so $a = 0.85$ and $C(t) = 80 \times 0.85^t$.
2. 3 h 30 min $= 3.5$ h. $C(3.5) = 80 \times 0.85^{3.5} \approx 45.3$ mg/L.
3. $45.3 < 50$: after 3 h 30 min the concentration has already dropped below 50 mg/L, so the new injection is due (in fact $C(3) \approx 49.1$ mg/L, so the concentration is below this threshold from 3 h onwards).

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 185 and the rate or ratio by 3. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.

Session 14 — Review and chapter assessment

Exponential Functions

12 Nov 2026 • B: 1 h – C: 2 h

Objectives.

- Use what the chapter has covered.
- Write up a modelling solution.
- Correct a mistake.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 3 - x + 2$. $2x + 5$
2. Is the function $x \mapsto 0.7^x$ increasing? **No, it is decreasing.**
3. Compare 1.04^3 and 1.04^7. $1.04^3 < 1.04^7$
4. Calculate $2^4 \times 2^3$. 2^7
5. Write 5^{-2} as a fraction. $\frac{1}{25}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have the key words rephrased: use what the chapter has covered and write up a modelling solution.

Activity 8. Using four pieces of student work, spot and correct: rate/multiplier confusion, a false power rule, an error in the direction of variation and a conclusion without a unit.

Production 1. A population of 2,000 inhabitants increases by 3% per year. Léa writes: "$P(n) = 2{,}000 \times 3^n$."

Production 2. Karim writes: "$2^x + 2^3 = 2^{x+3}$."

Production 3. Awa writes: "$f(x) = -3 \times 0.5^x$ is decreasing, because $0.5 < 1$."

Production 4. A machine bought for 12,000 euros loses 8% of its value per year. Tojo writes: "$V(n) = 12{,}000 \times 0.92^n$, so $V(5) \approx 7{,}909$. Answer: 7,909."

Solution. The correction must name the error, write the correct rule, redo the calculation and conclude in context. Student work 1: confusion between rate and multiplier; an increase of 3% corresponds to $q = 1.03$, so $P(n) = 2{,}000 \times 1.03^n$. Student work 2: false power property; the rule concerns products, $2^x \times 2^3 = 2^{x+3}$, but the sum $2^x + 2^3$ does not simplify (with $x = 1$: $2 + 8 = 10 \neq 16 = 2^4$). Student work 3: error in the direction of variation; 0.5^x is decreasing, but the coefficient -3 is negative and reverses the direction of variation, so f is increasing (for example $f(0) = -3 < f(1) = -1.5$). Student work 4: conclusion without unit; the model and the calculation are correct, but the conclusion must be given in context: after 5 years, the machine is worth about 7,909 euros. Have students say aloud which piece of data triggers the choice of method.

Written record. To validate an exponential model: identify V_0 and q, calculate with the units, check the order of magnitude and interpret the result.

Example 9. Take the data from the investigation and write up the solution in this order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The correction must name the error, write the correct rule, redo the calculation and conclude in context. Student work 1: confusion between rate and multiplier; an increase of 3% corresponds to $q = 1.03$, so $P(n) = 2{,}000 \times 1.03^n$. Student work 2: false power property; the rule concerns products, $2^x \times 2^3 = 2^{x+3}$, but the sum $2^x + 2^3$ does not simplify (with $x = 1$: $2 + 8 = 10 \neq 16 = 2^4$). Student work 3: error in the direction of variation; 0.5^x is decreasing, but the coefficient -3 is negative and reverses the direction of variation, so f is increasing (for example $f(0) = -3 < f(1) = -1.5$). Student work 4: conclusion without unit; the model and the calculation are correct, but the conclusion must be given in context: after 5 years, the machine is worth about 7,909 euros.

Exercise 2.8 · essential.

1. Simplify $3^{x+6} \div 3^x$ and $(2^x)^3$.
2. Study the direction of variation of $f(x) = 6 \times 1.06^x$ and $g(x) = -6 \times 0.8^x$.
3. An initial value of 1700 increases by 6% per year. Give the model and calculate the value after 6 years.

Solution. 3^6 and 2^{3x}. f is increasing because $1.06 > 1$ and $6 > 0$; 0.8^x is decreasing and the coefficient -6 is negative, so g is increasing. The model is $V(t) = 1700 \times \left(1 + \dfrac{6}{100}\right)^t$. After 6 years, $V(6) = 1700 \times 1.06^6 \approx 2{,}411.48$.

Application.

1. The price of a digital currency rises by 125% one year, then falls by 36% the following year. What is the average annual rate of change over these two years?
2. The price of a raw material rises by 80% one month, then falls by 55% the following month. What is the average monthly rate of change?
3. A student claims that, in question 2, the average rate is $\dfrac{80 - 55}{2} = 12.5\%$. Explain in a sentence why this is wrong.

Solution. **Target:** Average rate of two successive changes.

1. The multipliers are $1 + \dfrac{125}{100} = 2.25$ and $1 - \dfrac{36}{100} = 0.64$. The overall multiplier is $2.25 \times 0.64 = 1.44$. The average annual multiplier q satisfies $q^2 = 1.44$, so $q = \sqrt{1.44} = 1.2$: an average annual rise of 20%.
2. The multipliers are 1.8 and 0.45; the overall multiplier is $1.8 \times 0.45 = 0.81$. The average multiplier is $\sqrt{0.81} = 0.9$, that is an average monthly fall of 10%.
3. Successive changes combine by multiplying the multipliers, not by adding the rates:

the 55% fall applies to a price that has already risen. Over the two months the price is multiplied by 0.81 (an overall fall of 19%), which corresponds to an average fall of 10% per month, not a rise of 12.5%.

Homework. Rework question 3 of the essential exercise of the session with the following data: replace the initial value by 190 and the rate or ratio by 4. Write up in four lines: data, formula, calculation, conclusion.

Solution. The correction depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and give your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **1 h session:** 7 min of fluency; 10 min of pooling and written record; 35 min of essential exercises; 5 min of correction; 3 min of exit ticket.
2 h session: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast finishers to produce a counterexample, a justification or a second method.